

Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 1: Complex Numbers

Chapter 1: Complex Numbers

Why this matters

Complex numbers are the gateway skill of the whole Further Maths course. Almost every later chapter — Argand diagrams, roots of polynomials, even some vectors and matrices work — leans on being fluent with i , conjugates, and complex arithmetic. Get this chapter automatic and the rest of the course gets noticeably easier.

Concept: the imaginary unit

A quadratic like $x^2 + 1 = 0$ has no real solutions, because no real number squares to give -1 . We define a new number, i , with the property $i^2 = -1$, so that $x^2 + 1 = 0$ has solutions $x = \pm i$. A **complex number** is any number of the form $a + bi$, where a and b are real numbers. a is called the **real part** (written $\text{Re}(z)$) and b is called the **imaginary part** (written $\text{Im}(z)$). Note that b is a real number — $\text{Im}(z) = b$, not bi .

Key vocabulary

Term	Meaning
Complex number, z	$z = a + bi$, with a, b real
$\text{Re}(z)$	The real part, a
$\text{Im}(z)$	The imaginary part, b (a real number)
Conjugate, z^* (or $\bar{z}$)	$a - bi$, the "mirror image" of z
Purely imaginary	$a = 0$, so $z = bi$
Modulus, z	The distance of z from the origin (met fully in Chapter 2)

Concept: arithmetic with complex numbers

Addition and subtraction combine real parts with real parts and imaginary parts with imaginary parts, just like collecting like terms. Multiplication uses the distributive law exactly as with brackets in algebra, replacing i^2 with -1 wherever it appears.

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i$$

Worked example 1: basic arithmetic

EXAMPLE 1

Question. Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, find (a) $z_1 + z_2$, (b) $z_1 z_2$.

Step 1. Add real and imaginary parts separately: $z_1 + z_2 = (3 - 1) + (-2 + 4)i = 2 + 2i$. **(M1 A1)**

Step 2. Expand the product using brackets: $z_1 z_2 = (3 - 2i)(-1 + 4i) = -3 + 12i + 2i - 8i^2$. **(M1)**

Step 3. Replace i^2 with -1 and simplify: $-3 + 12i + 2i + 8 = 5 + 14i$. **(M1 A1)**

Answer: $z_1 + z_2 = 2 + 2i$, $z_1 z_2 = 5 + 14i$.

Exam tip

EXAM TIP

If you are shaky here, it is tempting to treat i like an unknown " x " and forget to simplify i^2 at the end. Get into the habit of circling every i^2 the moment it appears in an expansion, before you simplify anything else — it stops careless sign errors at the source.

Concept: conjugates and division

The conjugate of $z = a + bi$ is $z^* = a - bi$. Multiplying a complex number by its own conjugate always gives a real, non-negative result:

$$zz^* = (a + bi)(a - bi) = a^2 - (bi)^2 = a^2 + b^2$$

This is the key trick for dividing complex numbers: multiply numerator and denominator by the conjugate of the denominator, turning the denominator into a real number.

Worked example 2: dividing complex numbers

EXAMPLE 2

Question. Simplify $(5 + i) / (2 - 3i)$, giving your answer in the form $a + bi$.

Step 1. Multiply top and bottom by the conjugate of the denominator, $2 + 3i$: $(5 + i)(2 + 3i) / [(2 - 3i)(2 + 3i)]$. **(M1)**

Step 2. Expand the numerator: $(5 + i)(2 + 3i) = 10 + 15i + 2i + 3i^2 = 10 + 17i - 3 = 7 + 17i$. **(M1 A1)**

Step 3. Expand the denominator: $(2 - 3i)(2 + 3i) = 4 + 9 = 13$. **(A1)**

Step 4. Divide each part by 13: $(7 + 17i)/13 = 7/13 + (17/13)i$. **(A1)**

Answer: $7/13 + (17/13)i$.

Concept: solving quadratics with complex roots

When $b^2 - 4ac < 0$ in the quadratic formula, the square root of a negative number appears. Write $\sqrt{-k} = i\sqrt{k}$ for positive k , and proceed exactly as before. Complex roots of a quadratic with real coefficients always occur as a conjugate pair — this is a result you should be able to quote and use, and it will resurface in Chapter 4.

This happens because conjugating a real number changes nothing. If z is a root of an equation with real coefficients, taking the complex conjugate of the whole equation leaves every coefficient

unchanged but replaces z with z^* — so z^* must satisfy the same equation too, and is automatically also a root.

Worked example 3: quadratic with complex roots

EXAMPLE 3

Question. Solve $z^2 - 6z + 25 = 0$.

Step 1. Apply the quadratic formula: $z = [6 \pm \sqrt{(36 - 100)}] / 2$. **(M1)**

Step 2. Simplify the discriminant: $36 - 100 = -64$, so $\sqrt{(-64)} = 8i$. **(M1 A1)**

Step 3. $z = (6 \pm 8i)/2 = 3 \pm 4i$. **(A1)**

Answer: $z = 3 + 4i$ or $z = 3 - 4i$ (a conjugate pair, as expected for real coefficients).

Quick check: the sum of the roots should equal $-b/a$. Here $(3+4i) + (3-4i) = 6$, matching $-(-6)/1 = 6$. This same check works for any polynomial and is worth doing whenever time allows.

Concept: solving cubics and quartics with a given complex root

For a cubic or quartic with real coefficients, if one complex root is given, its conjugate is automatically also a root (see the box above). To find the remaining root(s):

1. Multiply $(z - \text{root})(z - \text{conjugate})$ to get a real quadratic factor.
2. Divide this quadratic into the original polynomial — by inspection, comparing coefficients, or algebraic long division — to find the remaining factor.
3. Solve the remaining factor for any real root(s).

Worked example 4: cubic with a complex root

EXAMPLE 4

Question. Given that $2 + i$ is a root of $z^3 - 7z^2 + 17z - 15 = 0$, find all three roots.

Step 1. State the conjugate root: $2 - i$ is also a root, since the coefficients are real. **(B1)**

Step 2. Form the quadratic factor from this conjugate pair: $(z - (2+i))(z - (2-i)) = (z - 2)^2 - i^2 = z^2 - 4z + 4 + 1 = z^2 - 4z + 5$. **(M1 A1)**

Step 3. Divide the cubic by this quadratic: $z^3 - 7z^2 + 17z - 15 = (z^2 - 4z + 5)(z - 3)$. Compare coefficients or use polynomial division to confirm the remaining linear factor is $(z - 3)$. **(M1 A1)**

Answer: The roots are $z = 2 + i$, $z = 2 - i$, $z = 3$.

Exam tip

EXAM TIP

Comparing coefficients (rather than long division) is usually faster and less error-prone here, because you already know the quadratic factor exactly. Write $z^3 - 7z^2 + 17z - 15 = (z^2 - 4z + 5)(z + a)$ and match the constant term first ($5a = -15$, so $a = -3$) — this is a quick check that also builds number sense for Chapter 4.

Chapter summary

Idea	Key point
Definition of i	$i^2 = -1$; $z = a + bi$ with a, b real
Arithmetic	Add/subtract like terms; multiply using brackets and simplify i^2
Conjugate	$z^* = a - bi$; $zz^* = a^2 + b^2$ is always real
Division	Multiply top and bottom by the conjugate of the denominator
Complex roots of quadratics	Occur in conjugate pairs when coefficients are real
Cubics/quartics	A given complex root gives its conjugate free; form the real quadratic factor and divide

Common exam mistakes

- Forgetting that $\text{Im}(z)$ is the real coefficient b , not bi .
- Sign errors when expanding $(a - bi)(a + bi)$ — forgetting that the middle terms cancel.
- Writing $\sqrt{-64} = -8i$ instead of the correct $8i$ (or losing the $\pm$ altogether).
- Assuming a complex root of a cubic automatically gives you the third root without doing the division — you must find the real root too.
- Not fully simplifying a division answer into $a + bi$ form (leaving a fraction unresolved or a fraction inside a fraction).
- Treating $\sqrt{-a} \times \sqrt{-b}$ as $\sqrt{ab}$ for positive a, b — the rule $\sqrt{x} \times \sqrt{y} = \sqrt{xy}$ only holds for non-negative reals. Convert each root to $i\sqrt{\text{positive number}}$ *before* multiplying: $\sqrt{-4} \times \sqrt{-9} = (2i)(3i) = 6i^2 = -6$, not $\sqrt{36} = 6$.

You should now be able to

- Add, subtract, multiply and divide complex numbers.
- Find and use the conjugate of a complex number.
- Solve any quadratic equation, including those with complex roots.
- Use a given complex root of a cubic or quartic (with real coefficients) to find all remaining roots.